Hesai AT512 for Emergency Brake Assist

Lidar performance – statistical modelling (Exercise 3)

Team: *names TBD*

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The lidar: Hesai AT512

range	300 m @ 10 % refl. (400 m max)
resolution	$0.05^\circ \times 0.05^\circ$
field of view	$120^\circ \times 25.6^\circ$
points	$12.3\,\mathrm{M/s} = 2400 \times 512$ @ 10 Hz

Why this one

- automotive-grade, long-range series product
- very fine angular grid → many beams on small, far objects
- 120° covers adjacent lanes

Assumption: the 300 m spec means 90 % detection probability (not stated by Hesai).

Model: returns per frame are a random variable

1. Beams on target (W × H object at R)

$$n = \frac{W}{R\,\Delta\theta_h} \cdot \frac{H}{R\,\Delta\theta_v}$$

random grid offset → n rounds up or down

2. Each beam returns with probability (echo $\propto \rho/R^2$)

$$p = 1 - e^{-C\rho/R^2}, \quad p(300\,\text{m}, 10\,\%) = 0.9$$

3. Returns: $K \sim \text{Binomial}(n, p)$
Requirement: $P(K \geq 5) \geq 99\,\%$ at 250 m

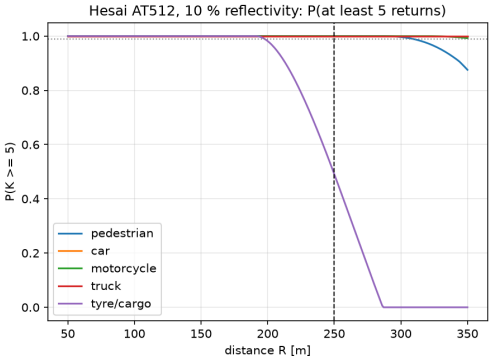
Example: pedestrian, 10 %, 350 m

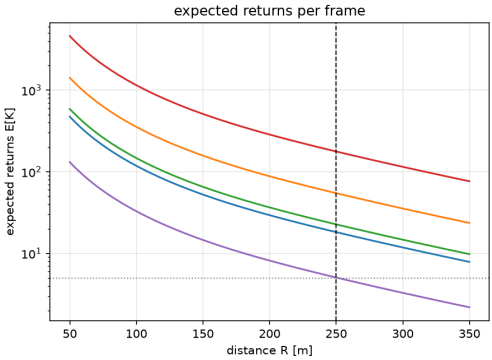
frame	1	2	3	4	5	6	7	8
beams	12	12	12	6	6	12	6	6
returns	11	10	10	5	5	10	**4**	**3**

average $E[K] = 7.9 \geq 5$
but $P(K < 5) = 12\,\%$ → **fails**

The average can pass while the probability fails.

Results (10 % reflectivity)





$P(K \geq 5)$ at 250 m: pedestrian and all vehicles $> 99.99\,\%$, tyre (0.5 m) **49 %**

Conclusion

AT512 fulfils EBA

- all vehicles and pedestrians at 250 m, even at 10 % reflectivity
- 99 % range: pedestrian 308 m, motorcycle 353 m, car/truck $\geq$ 400 m
- small debris ($\sim$0.5 m): only up to $\approx$ 200 m – too few beams hit (resolution limit, not reflectivity)

Limitations

- flat, frontal targets
- independent beams – optimistic in rain / dirty sensor
- 90 % behind the 300 m spec is assumed
- single frame; at 130 km/h, 34 frames remain before braking must start